1. Practical implementation

1.1. Stage I

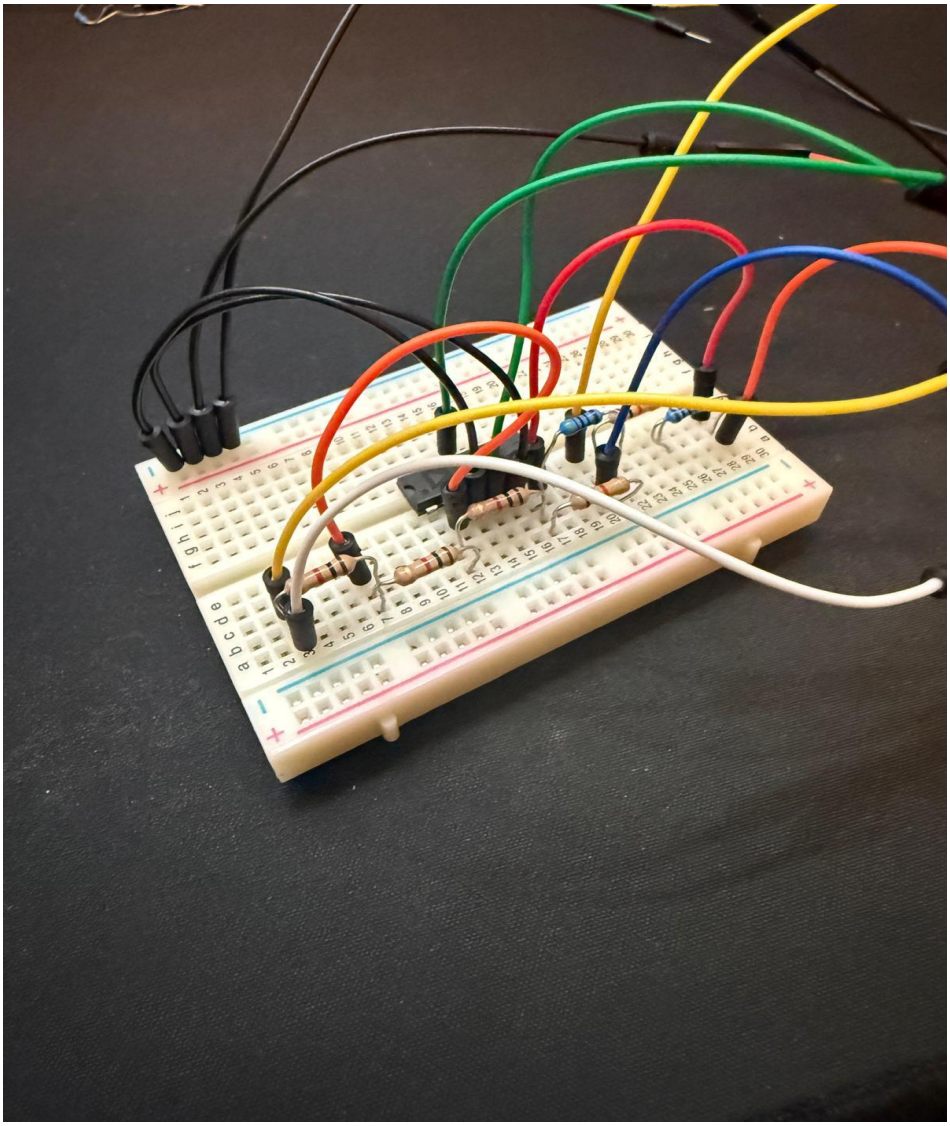

Figure 41. Stage I practical implementation

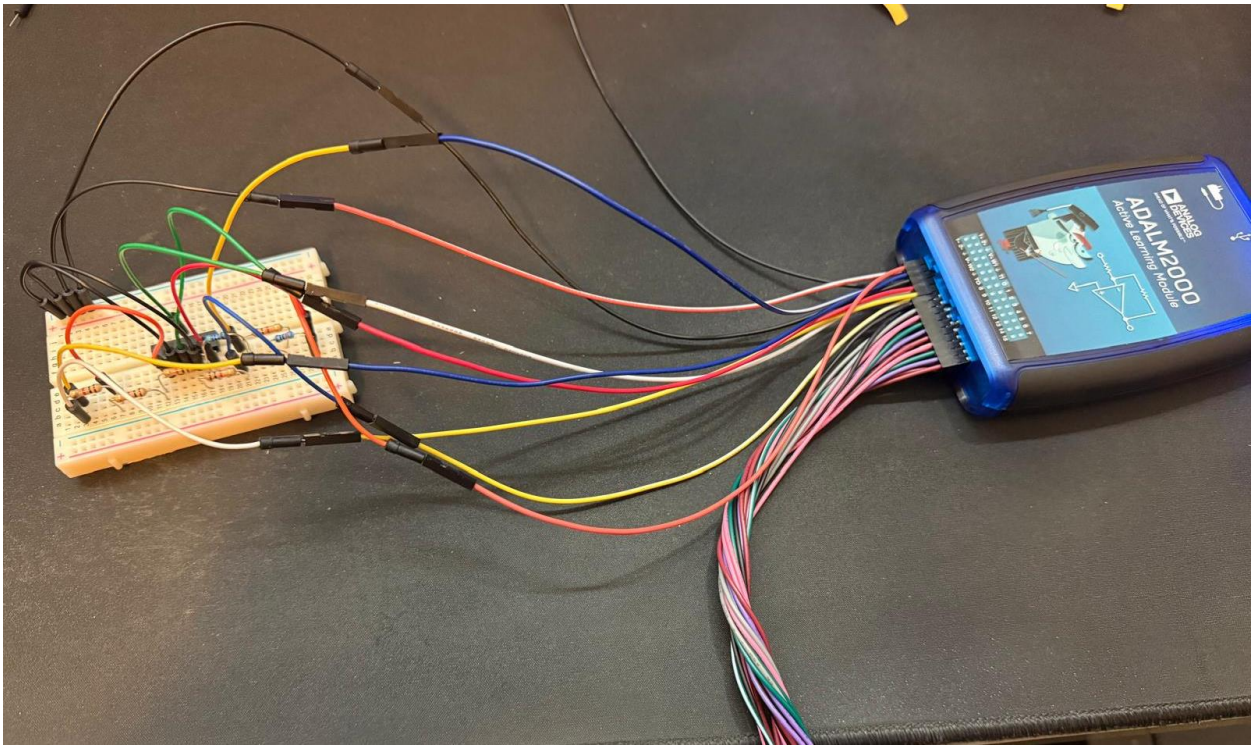

Figure 42. Stage I + ADALM2000 practical implementation

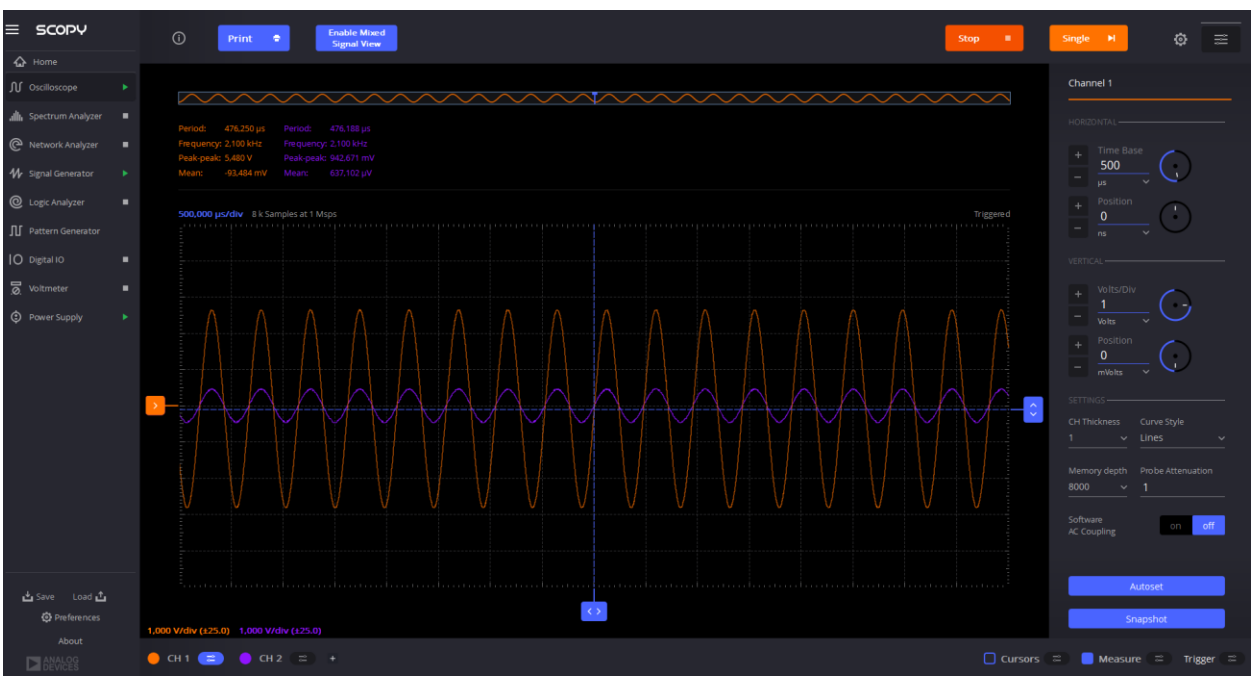

Figure 43. Output signal (orange) and input signal (purple) for A_{Vin} max

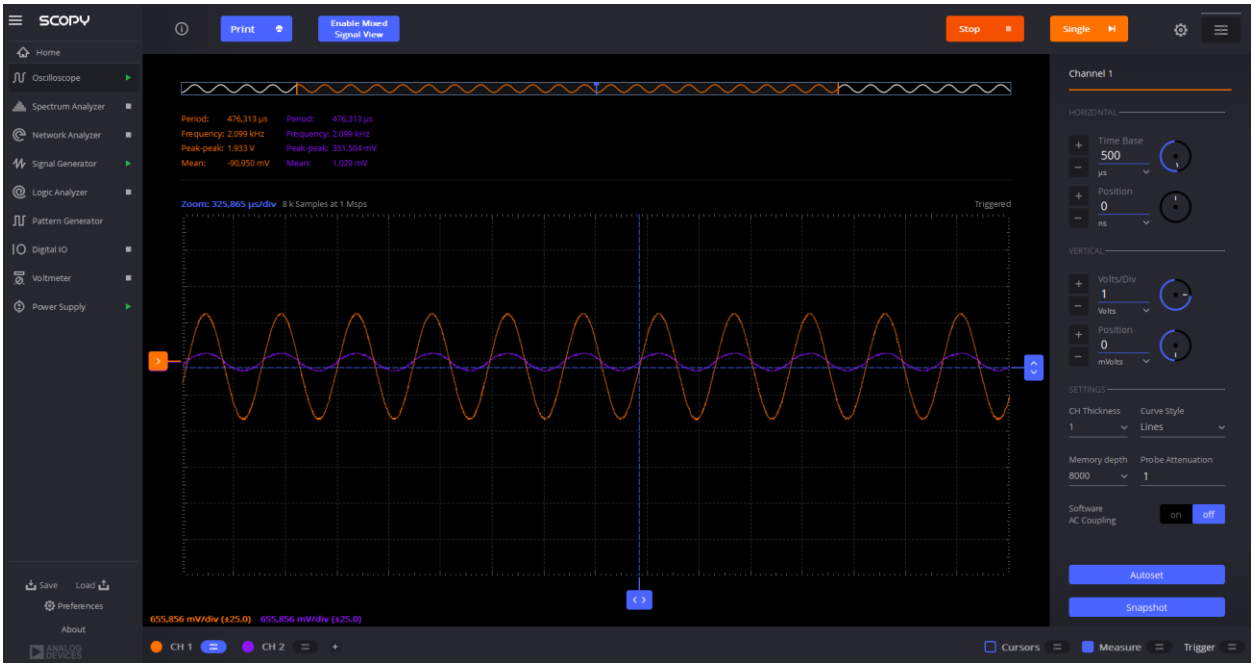

Figure 44. Output signal (orange) and input signal (puerple) for A_{Vin} min

Figure 45. Frequency analasys

From the obtained frequency characteristic we cannot determine the necessary parameters (gain and bandwidth) because it is not an expected Bode characteristic.

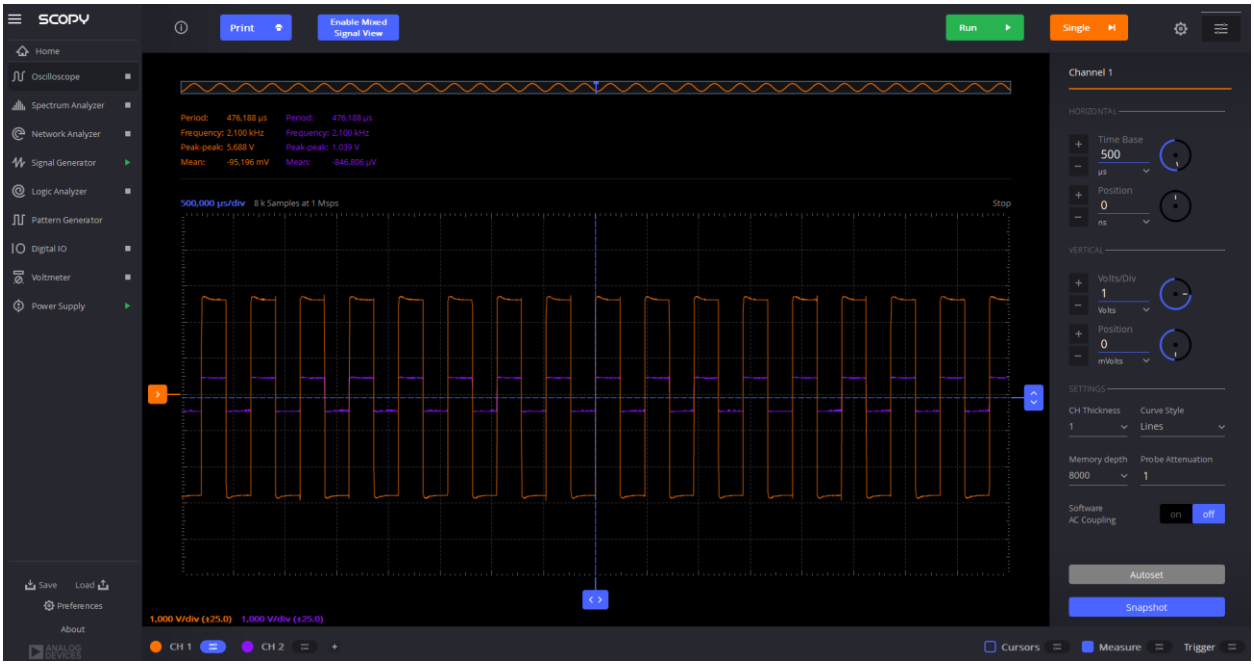

Figure 46. Output signal (orange) when rectangular signal at input (purple)

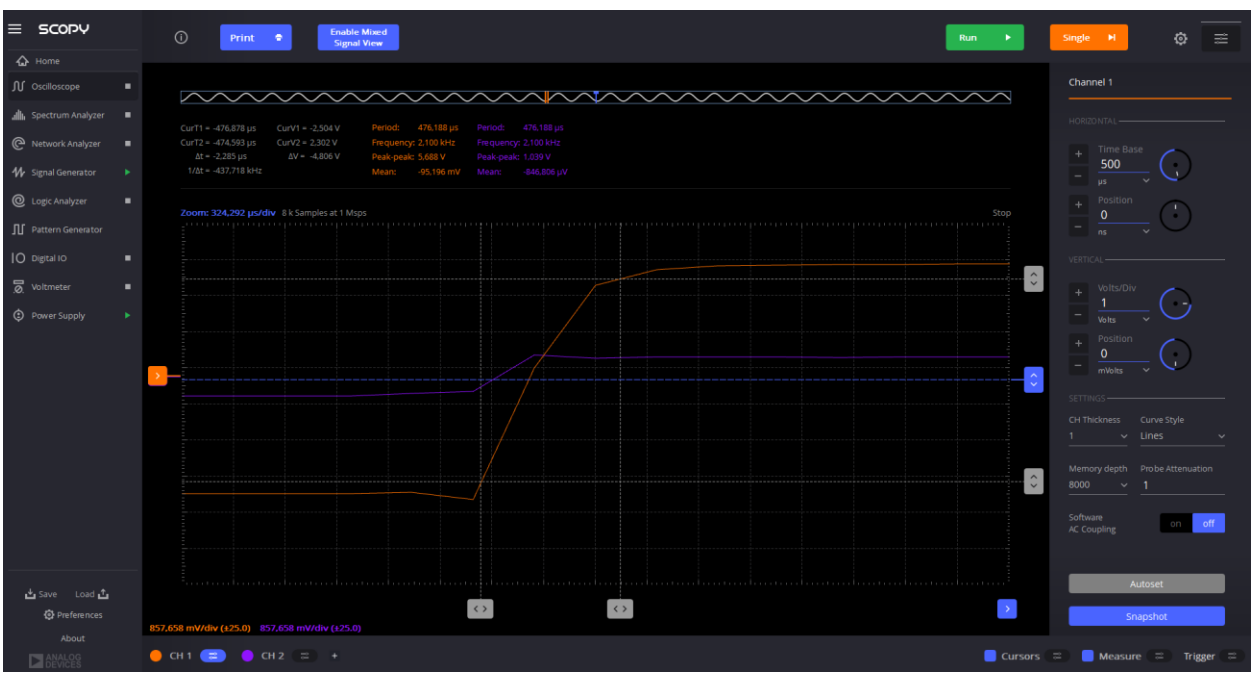

Figure 47. Slew Rate

The image above shows a Slew Rate=deltaV/deltaT=4.8V/2.28us=2.1V/us

1.2. Stage II

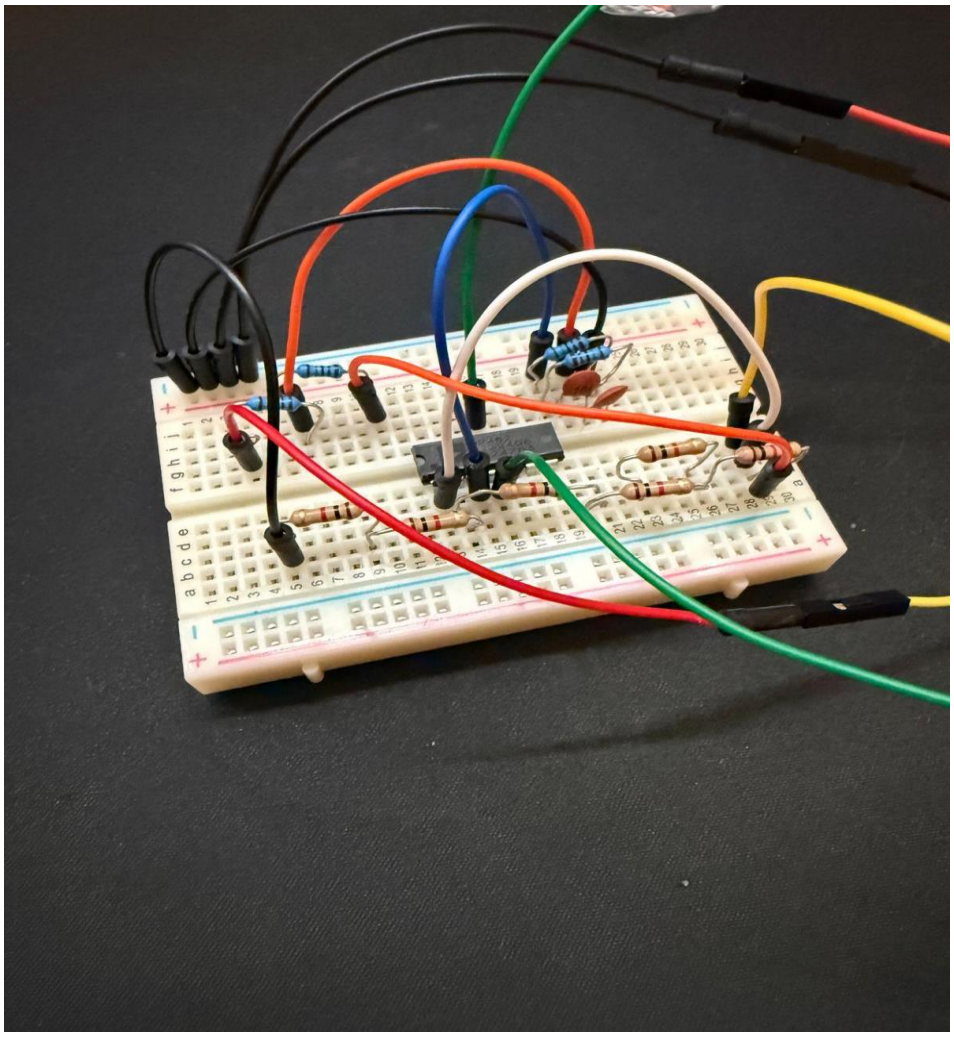

Figure 48. Stage II practical implementation

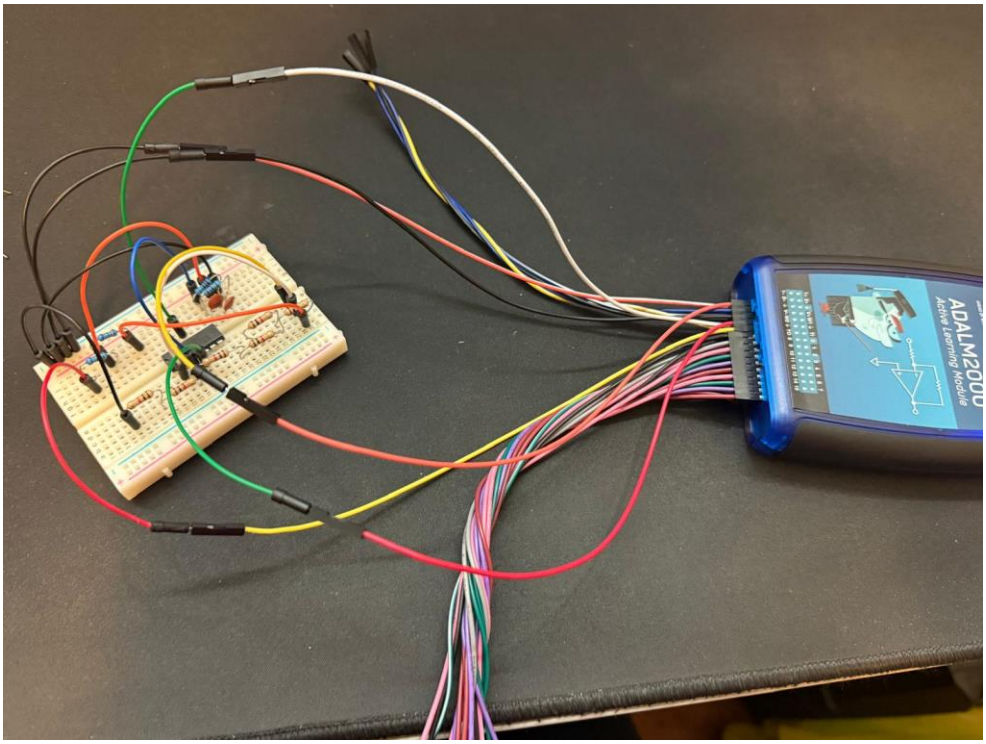

Figure 49. Stage II + ADALM2000 practical implementation

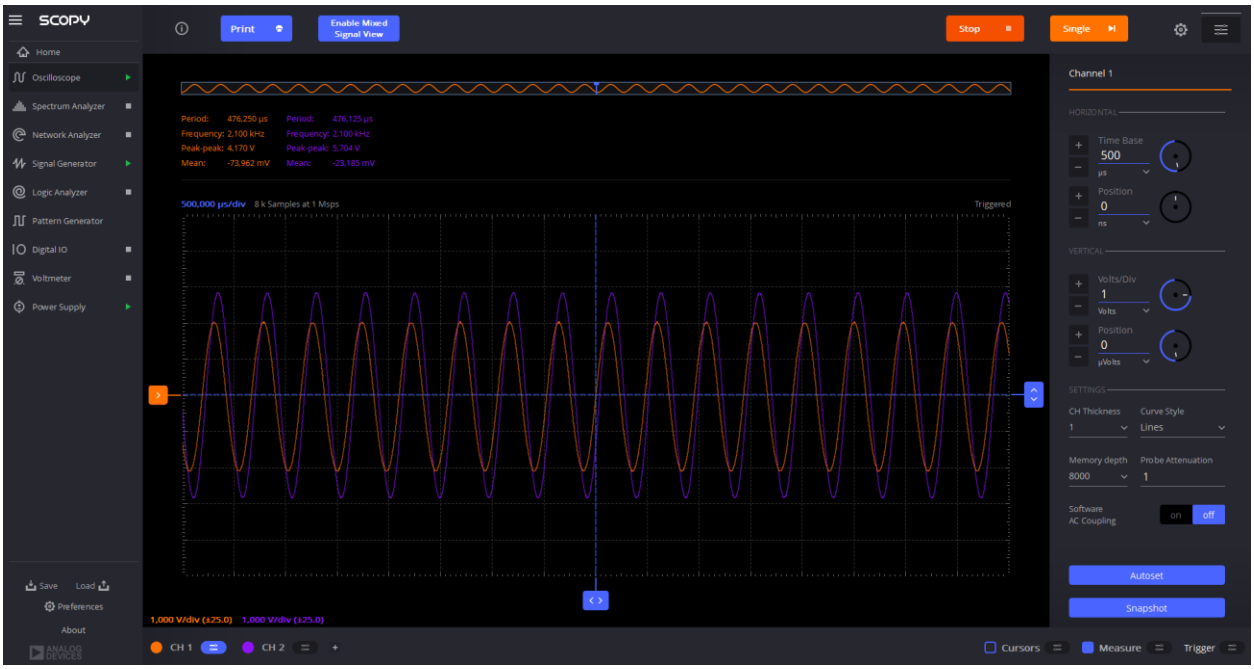

Figure 50. Output signal (orange) and input signal (purple) for A_{Vin} max

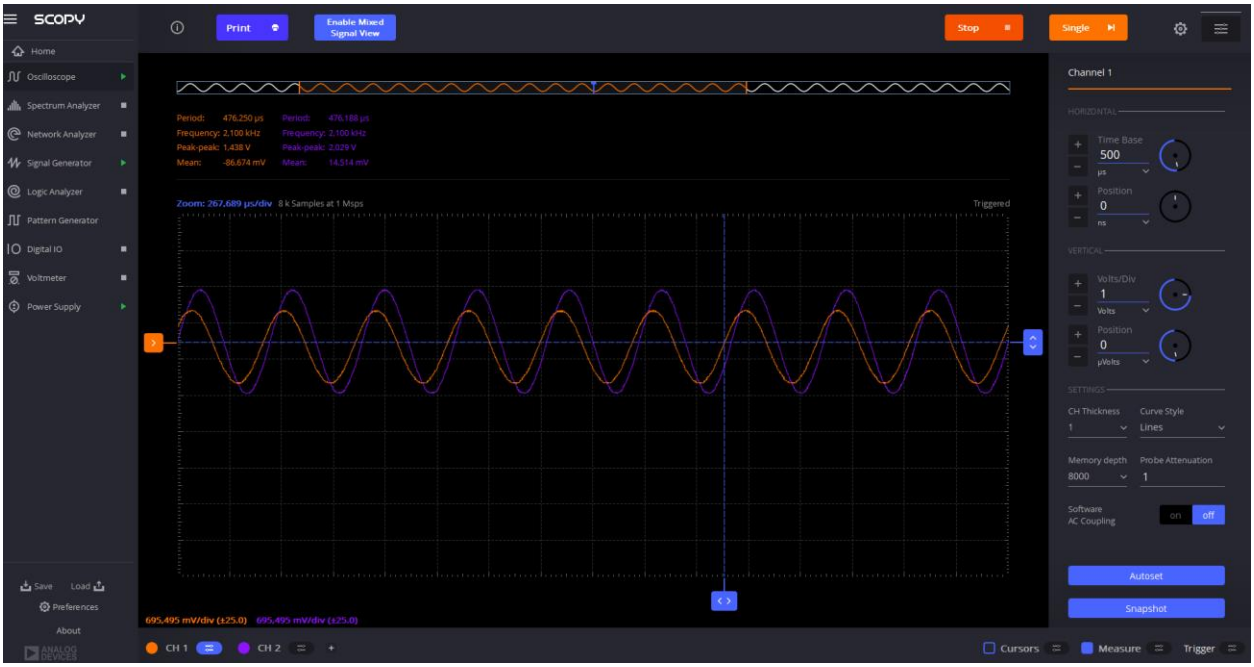

Figure 51. Output signal (orange) and input signal (purple) for A_{Vin} min

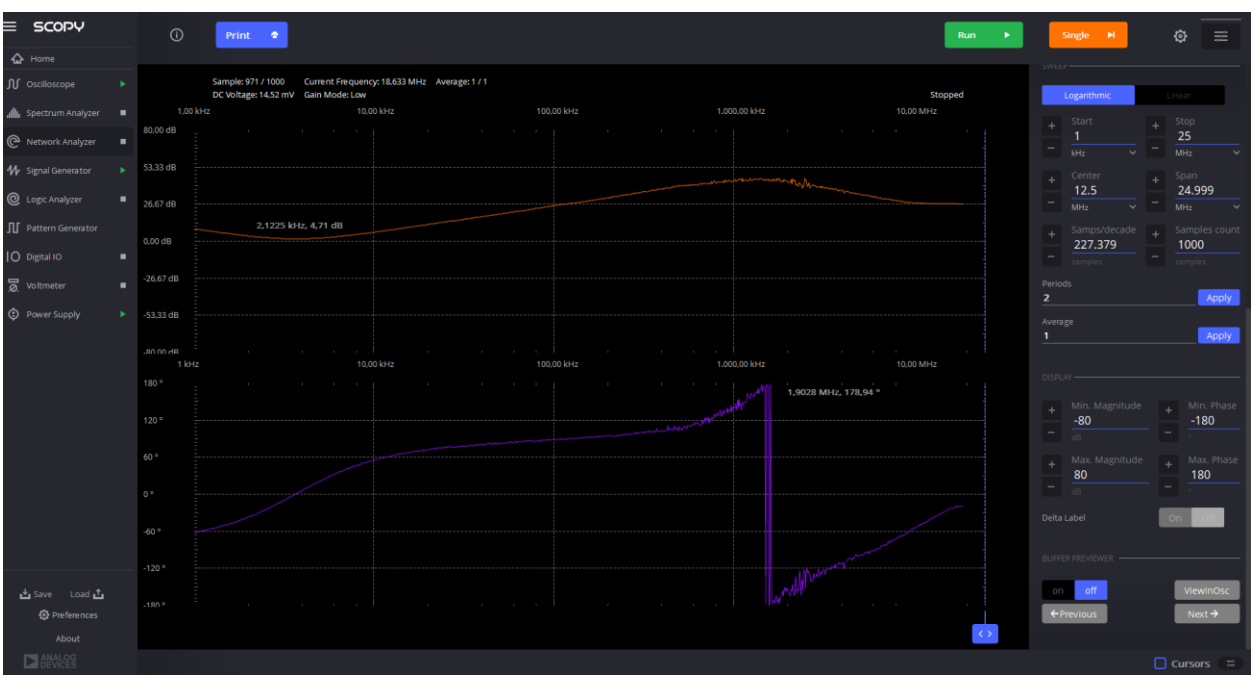

Figure 52. Frequency analasys

It is practically impossible to extract necesarry parameters because we did not obtain the wanted filter characteristic.

1.3. Stage III

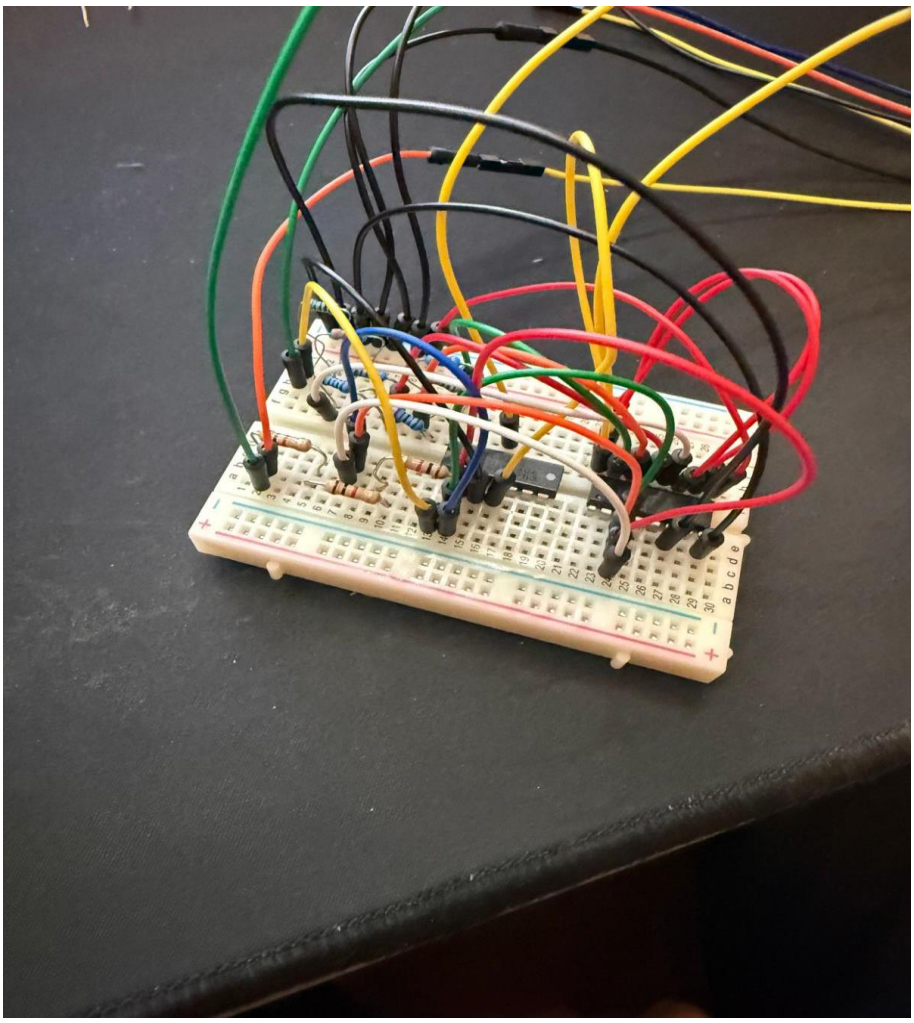

Figure 53. Stage III practical implementation

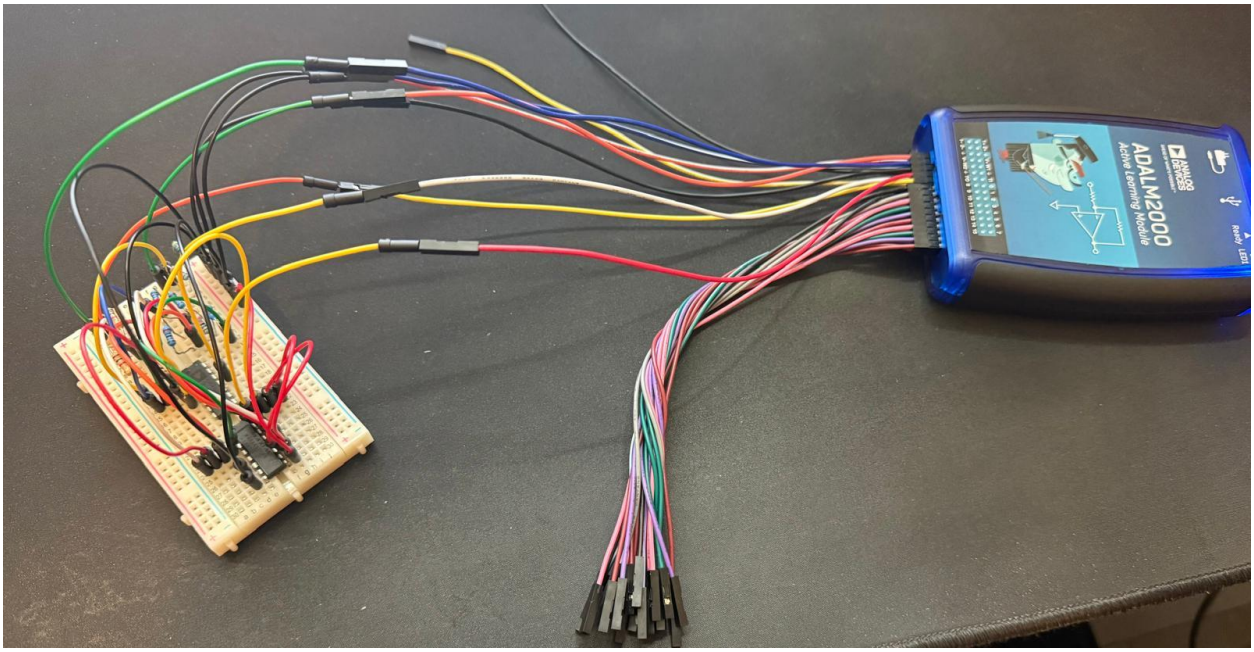

Figure 54. Stage III + ADALM2000 practical implementation

Figure 55. Output signal (orange) and input signal (purple) for A_{Vin} max and minimum PGA gain (2dB=1.25V/V)

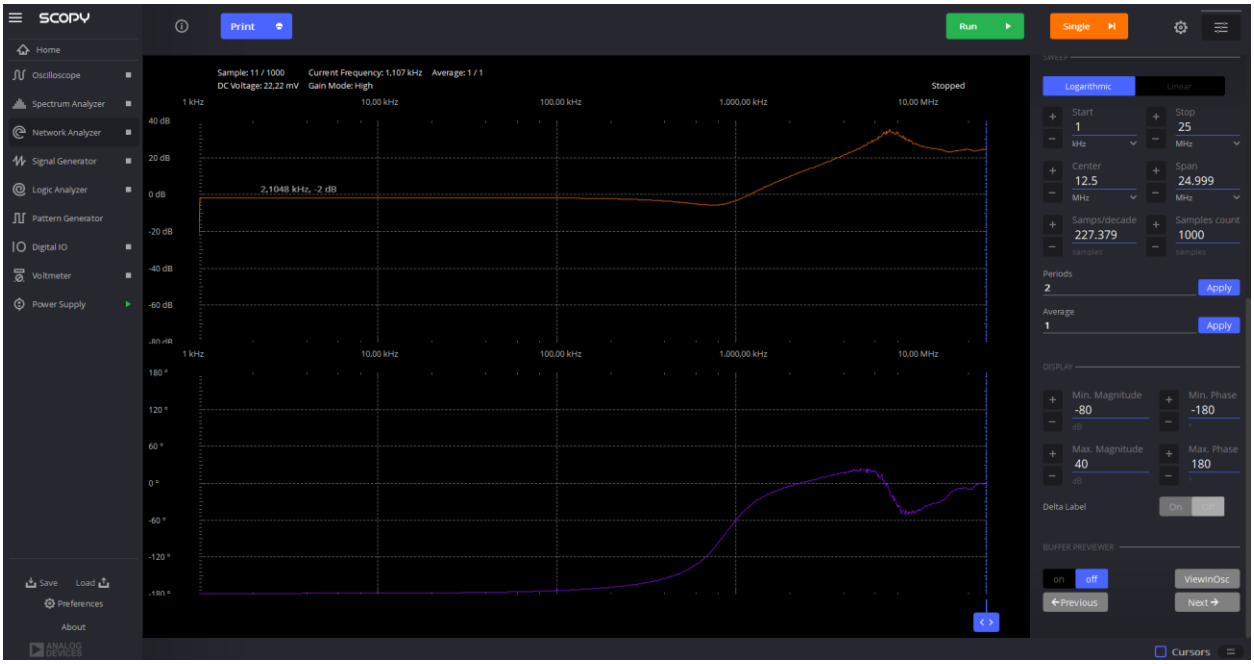

Figure 56. Frequency analysis for 2dB PGA gain

This characteristic does not help us correctly determine the 2dB gain of the PGA nor its bandwidth corresponding to this gain.

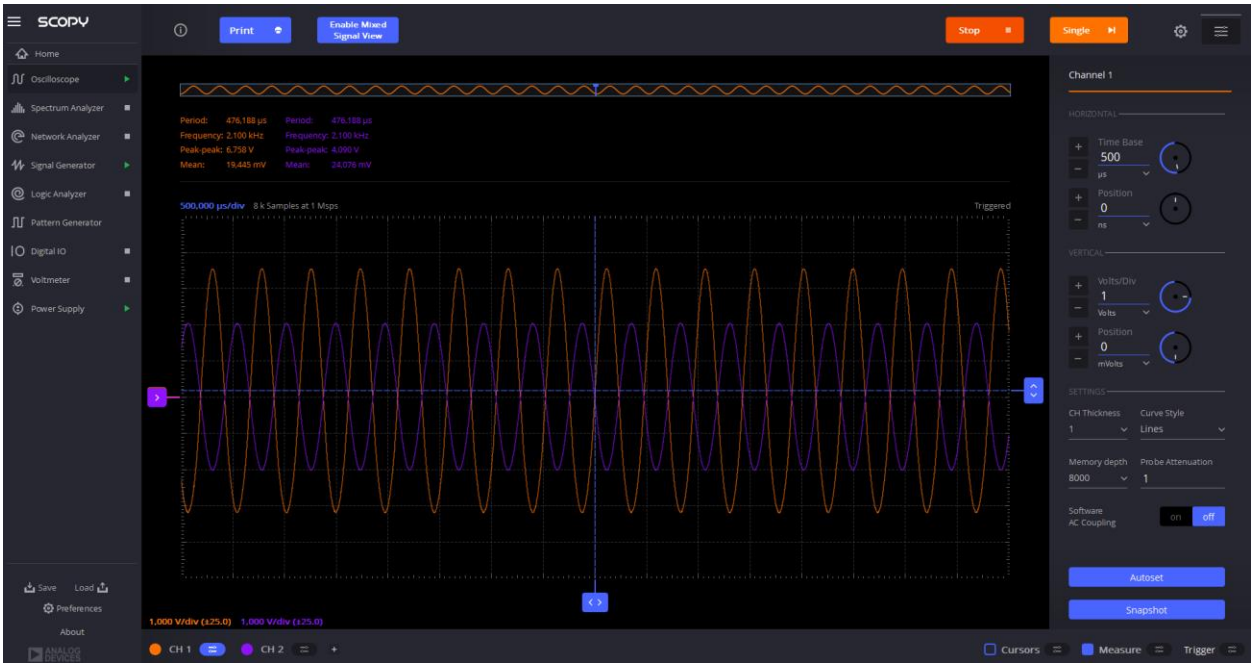

Figure 57. Output signal (orange) and input signal (purple)for A_{Vin}=4V peak-to-peak (slightly lower than A_{Vin} max) and PGA gain of 5dB=1.78V/V

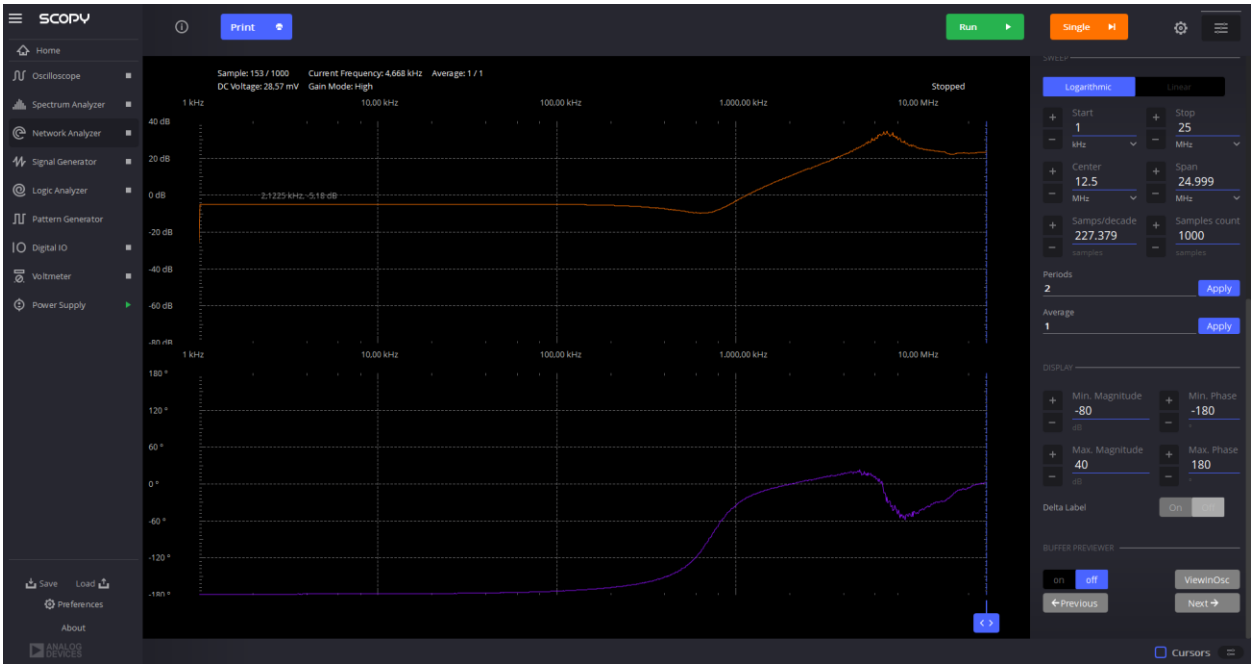

Figure 58. Frequency analysis for 5dB PGA gain

This characteristic does not help us correctly determine the 5dB gain of the PGA nor its bandwidth corresponding to this gain.

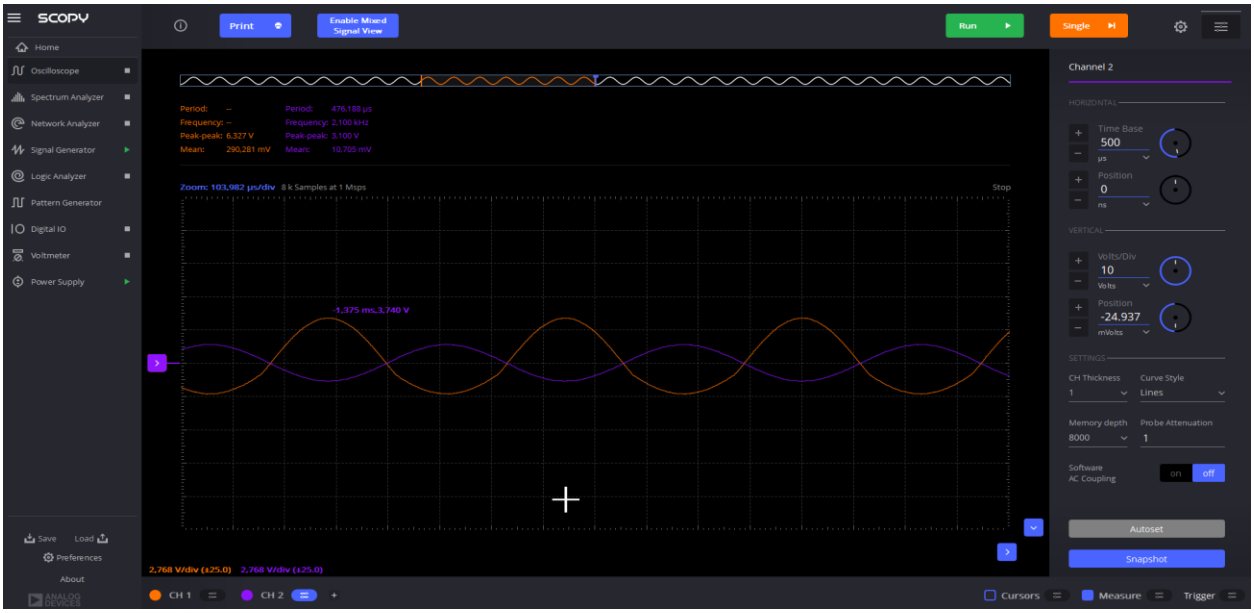

Figure 59. Output signal (orange) and input signal (purple) for A_{Vin}=3V peak-to-peak (slightly lower than A_{Vin} max) and PGA gain of 8dB=2.51V/V

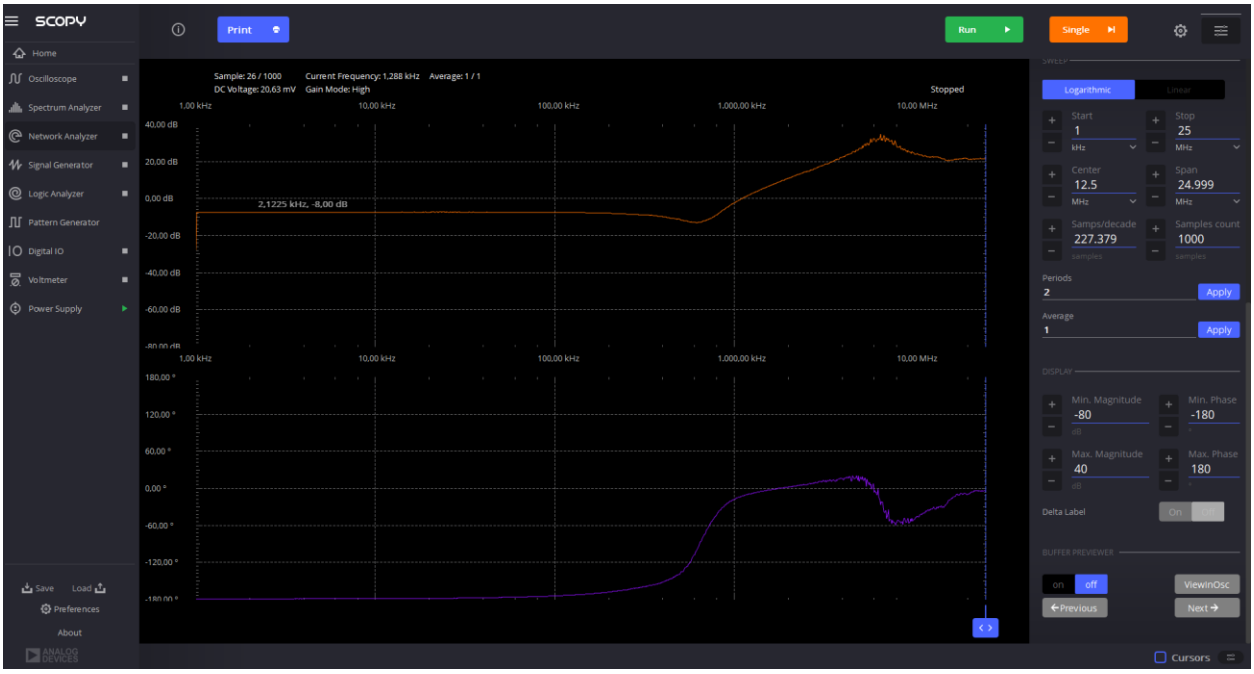

Figure 60. Frequency analysis for 8dB PGA gain

This characteristic does not help us correctly determine the 8dB gain of the PGA nor its bandwidth corresponding to this gain.

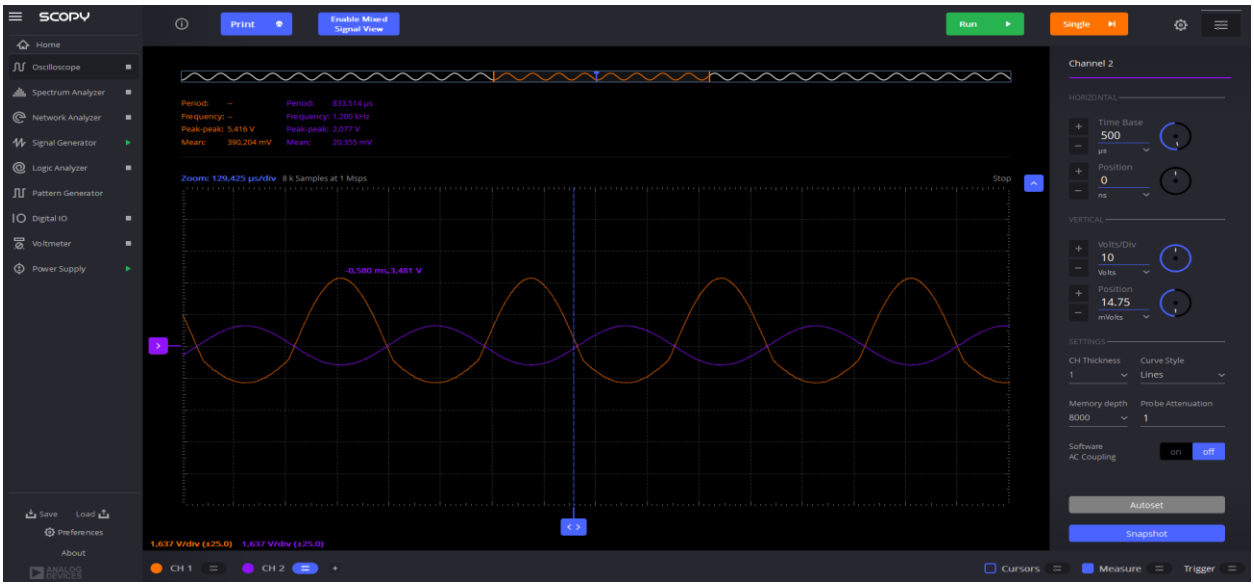

Figure 61. Output signal (orange) and input signal (purple) for A_{Vin} min and max PGA gain (11dB=3.55V/V)

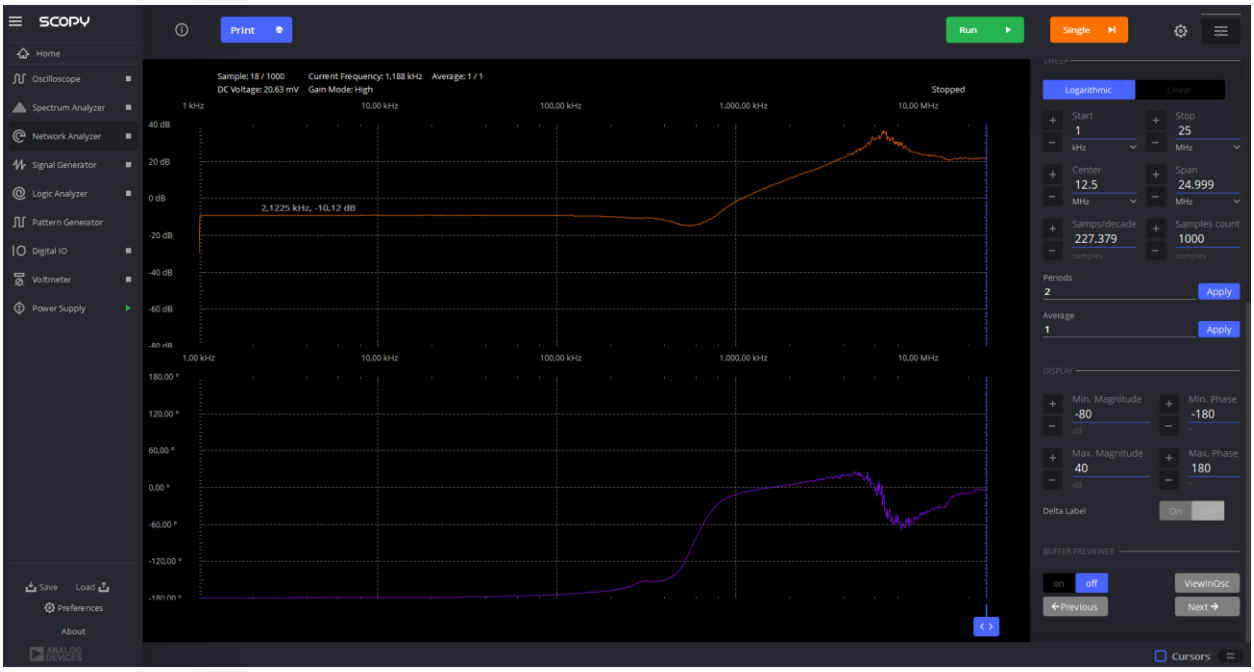

Figure 62. Frequency analysis for 11dB PGA gain

This characteristic does not help us correctly determine the 11dB gain of the PGA nor its bandwidth corresponding to this gain.

1.4. Stage IV

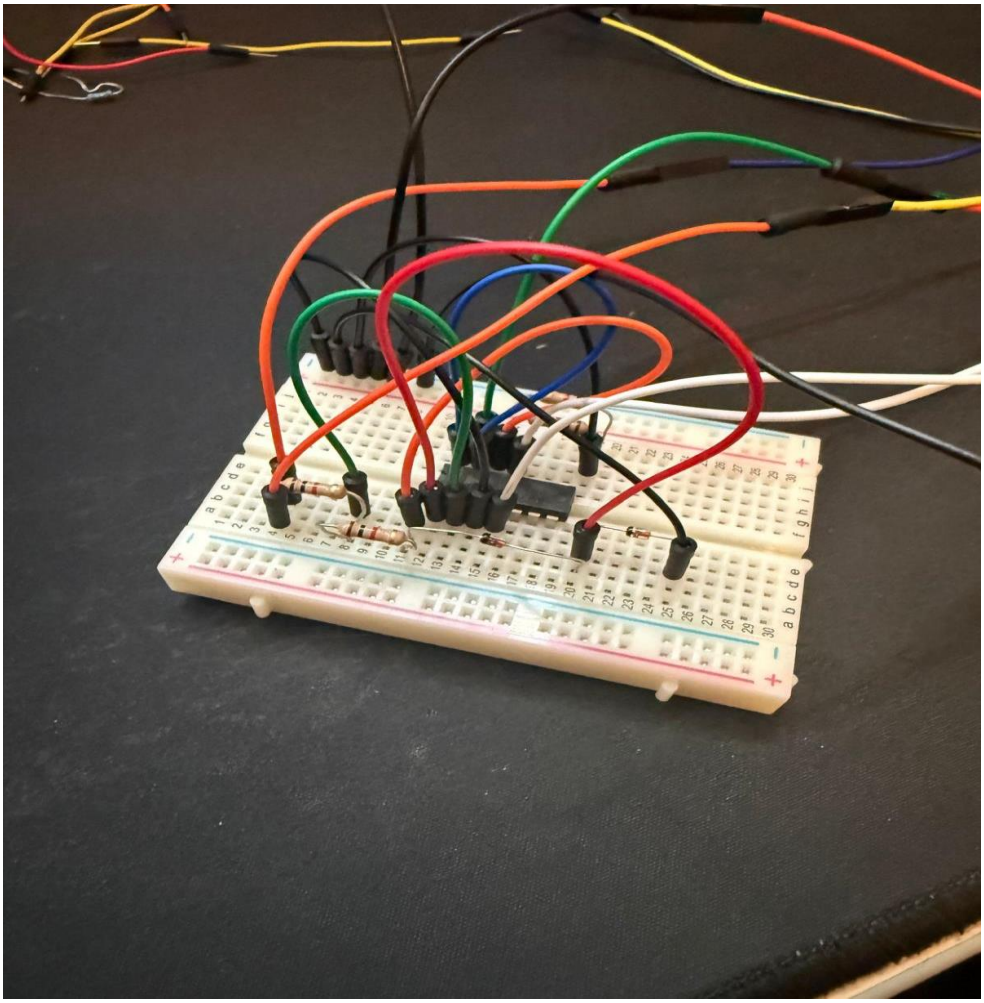

Figure 63. Stage IV practical implementation

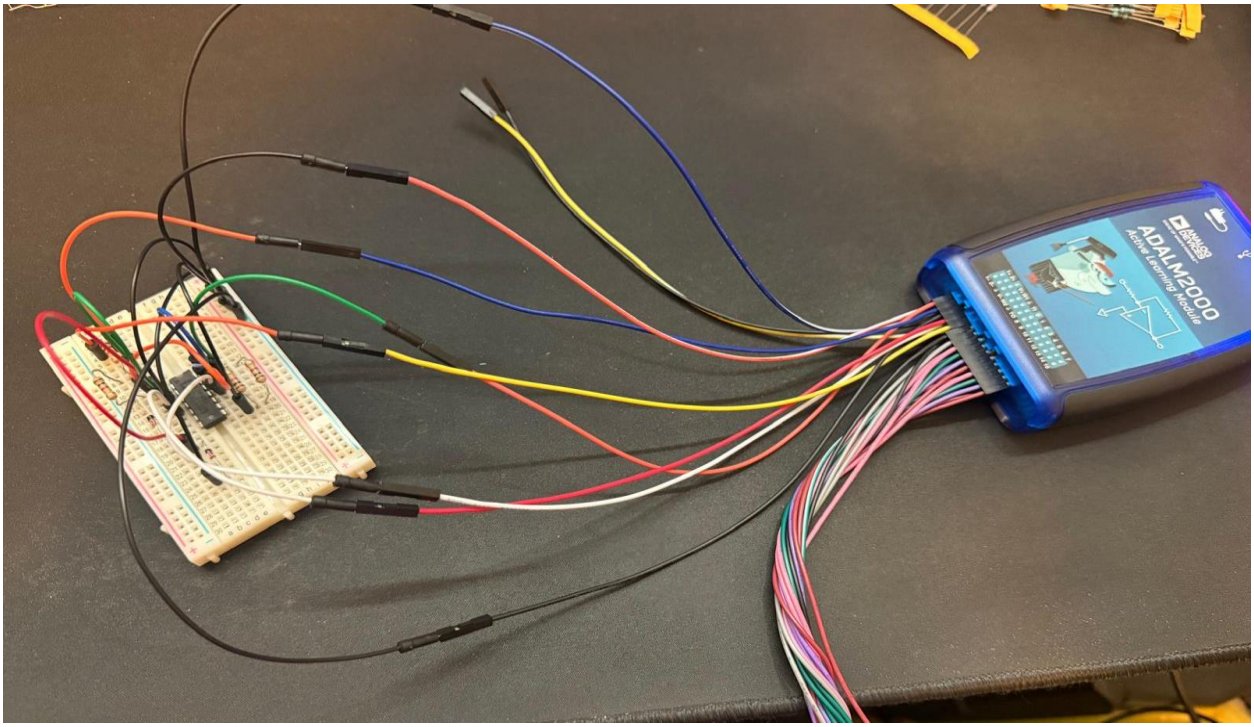

Figure 64. Stage IV + ADALM2000 practical implementation

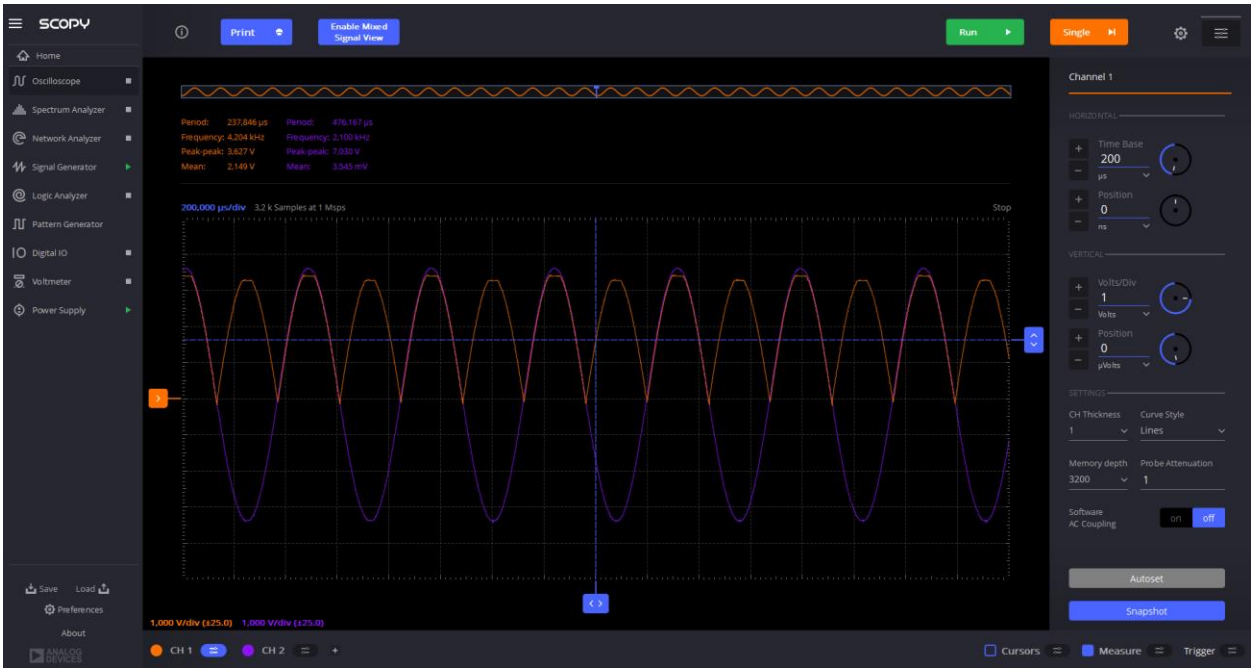

Figure 65. Output signal rectified (orange) and input signal (purple)

The signal from the rectifier`s output is limited at the top due to the power supply. A +/-5V supply is very limiting for a 3.6V amplitude of the output signal, and this is limited due to the Output Voltage Swing of the OP482 Op-Amp. The OP482 is not a Rail-to-RailOp-Amp, this means that it cannot reach the power supply ends, in this case +/-5V, it will go up to +/-3.5V according to the datasheet.

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- https://www.analog.com/media/en/technical-documentation/data-sheets/max4617-max4619.pdf
- **Resistor finder :** https://mustcalculate.com/electronics/resistorfinder.php?r=500&es=E6